

01

A 50-year-old patient presents to the clinic with progressive dysphagia to both solids and liquids over the past few months. The patient reports a sensation of food getting stuck in the chest and occasional regurgitation of undigested food. There is no significant weight loss or history of smoking. An esophagram shows a dilated esophagus with a narrow "bird-beak" appearance at the lower esophageal sphincter (LES). Further examination reveals failure of relaxation of the LES. What is the most likely diagnosis?

- ☐ Esophageal cancer
- ☒ Achalasia
- ☐ Gastroesophageal reflux disease (GERD)
- ☐ Esophageal stricture

**Explanation :**

The patient's symptoms of progressive dysphagia to both solids and liquids, along with the finding of a dilated esophagus and a "bird-beak" appearance on imaging, are classic for achalasia. Achalasia is a motility disorder characterized by the failure of the LES to relax and the absence of normal esophageal peristalsis. The classic radiologic finding is a dilated esophagus with a narrow LES, often described as a "bird-beak" appearance. This fits the patient's presentation of dysphagia to both solids and liquids.

While esophageal cancer can present with progressive dysphagia, it usually starts with difficulty swallowing solids and progresses to liquids. Additionally, weight loss and risk factors such as smoking or heavy alcohol use are often present.

02

A 55-year-old patient presents to the clinic with jaundice and pruritus. Laboratory tests reveal elevated serum bilirubin levels. An abdominal ultrasound shows a common bile duct (CBD) diameter of 10 mm and a hyperechoic lesion inside the CBD, suggesting a possible obstructive process. What is the next best step in management?

- ☐ IV antibiotics, maintenance fluids, and emergency surgery
- ☐ IV antibiotics, maintenance fluids, and an appointment after 6 months
- ☐ IV antibiotics, maintenance fluids, and analgesia
- ☒ ERCP (Endoscopic Retrograde Cholangiopancreatography)

**Explanation :**

The presence of jaundice, pruritus, elevated serum bilirubin, and an ultrasound finding of a dilated CBD with a hyperechoic lesion strongly suggests an obstructive process, most likely due to a CBD stone (choledocholithiasis). ERCP is a minimally invasive procedure that allows for both the diagnosis and treatment of biliary obstructions. It can be used to remove stones from the CBD, relieving the obstruction and addressing the patient's symptoms.

Delaying definitive treatment for an obstructive process can lead to complications such as cholangitis or pancreatitis. Immediate intervention is necessary. While emergency surgery may be necessary in some cases of complicated biliary obstruction (such as cholangitis), the first-line intervention for suspected choledocholithiasis is typically less invasive. ERCP is preferred for both diagnosis and treatment.

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diagnosis and treatment.

03

A 55-year-old female patient underwent breast surgery for stage IIIC breast cancer three months ago. The biopsy results reveal ductal carcinoma with metastatic involvement of the lymph nodes. She has completed her surgical treatment and is now in your office for follow-up. What should be your next step in management?

- ☐ Follow up for recurrence
- ☒ Refer for hormone and chemotherapy
- ☐ Reassure she is cured
- ☐ Schedule a mammogram of the other breast after 3 months

**Explanation :**

Given the patient's stage IIIC breast cancer with lymph node involvement, further treatment beyond surgery is typically required. The goal is to reduce the risk of recurrence and address any potential microscopic metastatic disease. For stage IIIC breast cancer, adjuvant chemotherapy, hormone therapy (if the tumor is hormone receptor-positive), and sometimes radiation therapy are indicated to reduce the risk of recurrence and to treat any microscopic metastatic disease. Multimodal therapy is crucial for improving the patient's prognosis.

04

A 45-year-old female patient with a known history of HIV presents with a one-week history of left breast pain and a palpable mass. On physical examination, a tender mass is felt in the left breast. Ultrasound reveals a 10x11 cm cyst with internal echogenic debris, suggestive of an abscess. The patient is afebrile and has stable vital signs. What is the best management for this patient?

- ☐ Open surgery and antibiotics
- ☒ Catheter drainage, antibiotics, and analgesia
- ☐ Incision and drainage, plus analgesia
- ☐ Observation and follow-up in one week

**Explanation :**

For a patient with an abscess, especially of significant size, and considering the patient's HIV status which might compromise healing, the appropriate management involves drainage and antibiotics. Catheter drainage (percutaneous drainage) is a minimally invasive procedure that effectively drains the abscess and is often guided by ultrasound. This approach is appropriate given the size of the cyst and the presence of internal debris. Antibiotics are necessary to treat any underlying infection, and analgesia will manage the patient's pain.

Open surgery is generally reserved for abscesses that are not amenable to less invasive procedures or have not responded to initial drainage attempts. It's more invasive than necessary for a first-line treatment.

Observation without intervention is inappropriate for a symptomatic abscess, especially one of this size. Delay in treatment can lead to worsening infection and complications.

05

A 35-year-old male patient presents to the emergency department with acute chest pain of one day duration. The pain is pleuritic in nature. Vital signs are normal except for tachycardia. An ECG





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05

A 35-year-old male patient presents to the emergency department with acute chest pain of one day duration. The pain is pleuritic in nature. Vital signs are normal except for tachycardia. An ECG shows diffuse ST elevation in all leads. What is the most likely diagnosis?

- ☐ Stable Angina
- ☐ Acute Coronary Syndrome
- ☒ Acute Pericarditis
- ☐ Pulmonary Embolism

**Explanation :**

The clinical presentation and ECG findings strongly suggest acute pericarditis. Acute pericarditis often presents with pleuritic chest pain, which can worsen with inspiration or when lying flat, and can be relieved by sitting up and leaning forward. The hallmark ECG finding in acute pericarditis is diffuse ST elevation in multiple leads, often accompanied by PR segment depression. While myocardial infarction can present with chest pain and ST elevation, the ST elevation in myocardial infarction is typically localized to specific leads based on the area of the heart affected. Diffuse ST elevation in all leads is not characteristic of myocardial infarction. Pulmonary embolism can cause pleuritic chest pain and tachycardia, but it does not typically cause diffuse ST elevation on an ECG. ECG findings in pulmonary embolism may include sinus tachycardia, right ventricular strain pattern, or the classic S1Q3T3 pattern.

06

What is the most likely causative agent for the above disease?

- ☒ Virus
- ☐ Bacteria
- ☐ Tuberculosis (TB)
- ☐ Malignancy

**Explanation :**

Acute pericarditis, which is characterized by pleuritic chest pain and diffuse ST elevation on ECG, is most commonly caused by viral infections. Viral infections are the most common cause of acute pericarditis. Common viral etiologies include Coxsackievirus, Echovirus, and other enteroviruses.

07

A 60-year-old male patient with a known history of hypertension and diabetes presents to the emergency department with left-sided chest pain that radiates to the shoulder. The pain has been persistent for the past hour. The patient describes the pain as pressure-like and it is not relieved by rest or nitroglycerin. There are no recent respiratory infections or signs of systemic illness. What is the most likely diagnosis?

- ☒ Acute Coronary Syndrome
- ☐ Pericarditis
- ☐ Musculoskeletal pain
- ☐ Gastroesophageal reflux disease (GERD)

**Explanation :**



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- ☒ Acute Coronary Syndrome
- ☐ Pericarditis
- ☐ Musculoskeletal pain
- ☐ Gastroesophageal reflux disease (GERD)

**Explanation :**

Given the patient's history of hypertension and diabetes, and the nature of the chest pain, ACS is the most likely diagnosis. ACS encompasses conditions such as unstable angina and myocardial infarction, which are common in patients with risk factors like hypertension and diabetes. The description of left-sided chest pain radiating to the shoulder that persists and is not relieved by rest or nitroglycerin is classic for ACS.

While musculoskeletal pain can cause chest pain, it is typically localized and reproducible with movement or palpation. It is less likely to radiate to the shoulder and more likely to be relieved by rest or changes in position.

GERD can cause chest pain, but it is usually described as a burning sensation and may be associated with regurgitation or a sour taste in the mouth. The pain is often related to meals and may be relieved by antacids, which is not described in this case.

08

A 1-month-old neonate is brought to the clinic due to failure to suck, inactivity, and sluggishness. The parents also report that the baby has cold skin. The baby has no significant birth history and was delivered full-term. On physical examination, the baby is noted to be lethargic with a slow heart rate and cool extremities. What is the most likely diagnosis?

- ☐ Sepsis
- ☒ Hypothyroidism
- ☐ Jaundice
- ☐ Congenital heart disease

**Explanation :**

The symptoms of failure to suck, inactivity, sluggishness, and cold skin in a neonate are highly suggestive of congenital hypothyroidism, which is a critical condition to diagnose and treat early to prevent developmental delays and other complications. Congenital hypothyroidism can present with symptoms like poor feeding, lethargy, a hoarse cry, cold skin, and a slow heart rate. Early diagnosis and treatment are essential to prevent intellectual disability and growth failure.

Sepsis can present with lethargy and poor feeding in neonates, it typically also includes signs of infection such as fever or hypothermia, irritability, and respiratory distress. The presentation of cold skin and sluggishness without other signs of infection is less consistent with sepsis.

Jaundice in neonates primarily causes yellowing of the skin and sclera due to elevated bilirubin levels. While severe jaundice can cause lethargy, the presentation of cold skin and failure to suck is more characteristic of hypothyroidism.



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A 55-year-old male patient with a history of chronic kidney disease (CKD) and hypertension presents for a routine follow-up. His most recent laboratory results show a creatinine level of 2.5 mg/dL and an estimated glomerular filtration rate (eGFR) of 40 mL/min/1.73 m<sup>2</sup>. He has been compliant with his antihypertensive medications, which include an ACE inhibitor and a diuretic. You are discussing his target blood pressure (BP) goals to manage his CKD and hypertension effectively. What is the target blood pressure in a patient with chronic kidney disease related to hypertension?

- ☐ <120/80 mmHg
- ☒ <130/80 mmHg
- ☐ <140/90 mmHg
- ☐ <150/90 mmHg

**Explanation :**

The target blood pressure for patients with chronic kidney disease (CKD) and hypertension is <130/80 mmHg to help slow the progression of kidney damage and reduce cardiovascular risk. According to guidelines, the recommended target blood pressure for patients with CKD and hypertension is <130/80 mmHg. Achieving this target helps to slow the progression of CKD and reduce the risk of cardiovascular events.

10

A 25-year-old female presents to the emergency department after falling from a tree. She complains of severe pain, significant swelling, and extensive bruising over her left elbow area. On examination, there is obvious deformity with angulation of the elbow, and the patient is unable to move her arm due to pain. What is the most appropriate management principle?

- ☒ Open reduction, followed by internal fixation
- ☐ Closed reduction, followed by immobilization
- ☐ Splinting and referral for orthopedic evaluation
- ☐ Observation and analgesia, with reassessment after 24 hours

**Explanation :**

The most appropriate management principle is open reduction followed by internal fixation. This involves surgical intervention to directly visualize the fractured bones, realign them to their anatomical position, and stabilize them with hardware such as plates, screws, or pins. Open reduction is necessary when there is significant displacement or complex fracture patterns that cannot be adequately managed with closed reduction alone. Internal fixation helps maintain proper alignment of the fractured bones and facilitates optimal healing. Closed reduction involves manipulation of the fractured bones back into alignment without surgical exposure of the fracture site. However, in cases of significant deformity and inability to achieve adequate reduction, open reduction is preferred. Splinting may provide temporary stabilization, it is not sufficient as definitive treatment for a complex elbow fracture-dislocation with significant deformity. Prompt surgical intervention is necessary to achieve proper alignment and prevent long-term complications.

11

A 10-year-old child presents to the pediatric clinic with complaints of persistent oozing discharge and pain on the lower leg. The child's parents report a history of trauma to the mid-anterior aspect





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A 10-year-old child presents to the pediatric clinic with complaints of persistent oozing discharge and pain on the lower leg. The child's parents report a history of trauma to the mid-anterior aspect of the leg three months ago, which resulted in a fracture. Since then, the child has experienced intermittent pain and swelling in the affected area. Over the past week, the child has developed worsening pain, along with the appearance of purulent discharge and redness over the fracture site. On examination, there is tenderness to palpation, erythema, warmth, and swelling over the mid-anterior aspect of the leg. What is the most likely diagnosis?

- ☐ Septic arthritis
- ☒ Osteomyelitis
- ☐ Cellulitis
- ☐ Compartment syndrome

**Explanation :**

Given the detailed history of trauma to the mid-anterior leg resulting in a fracture three months ago, coupled with the progressive symptoms of persistent pain, purulent discharge, redness, tenderness, erythema, warmth, and swelling over the fracture site, the most likely diagnosis is osteomyelitis. Osteomyelitis is characterized by infection of the bone, often following trauma or surgery. It presents with local signs of inflammation such as pain, tenderness, erythema, warmth, and possibly discharge. Given the history of trauma and the clinical presentation, osteomyelitis is the most likely diagnosis.

Septic arthritis involves infection of a joint space, typically presenting with acute joint pain, swelling, and limited range of motion. The presentation described here, with symptoms localized to the bone and surrounding tissues, is more consistent with osteomyelitis.

Compartment syndrome involves increased pressure within a closed fascial compartment, leading to compromised blood flow and tissue ischemia. While it can occur following trauma, the presentation here is more suggestive of an infectious process rather than compartment syndrome.

12

A 40-year-old male presents to the emergency department with sudden-onset severe epigastric pain that awakened him from sleep. He describes the pain as sharp, stabbing, and located centrally, radiating to his back. He denies any recent history of heavy alcohol consumption or significant dietary indiscretions. On examination, he appears diaphoretic and in significant distress. His abdomen is tender to palpation throughout, with voluntary guarding and rebound tenderness noted. Bowel sounds are diminished. What is the most likely diagnosis?

- ☒ Perforated peptic ulcer disease (PUD)
- ☐ Acute pancreatitis
- ☐ Perforated appendicitis
- ☐ Biliary colic

**Explanation :**

Perforation of a peptic ulcer can lead to the leakage of gastric contents into the peritoneal cavity, causing severe abdominal pain and signs of peritoneal irritation such as guarding and rebound tenderness. The patient's presentation is consistent with this diagnosis. The patient's sudden-onset severe epigastric pain, radiation to the back, and signs of peritoneal irritation are highly suggestive of gastric perforation, which is a surgical emergency requiring prompt intervention.

Perforated appendicitis typically presents with localized right lower quadrant pain that may later become diffuse. While it can cause peritoneal irritation, it is less likely to present with severe





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**Explanation :**

Perforation of a peptic ulcer can lead to the leakage of gastric contents into the peritoneal cavity, causing severe abdominal pain and signs of peritoneal irritation such as guarding and rebound tenderness. The patient's presentation is consistent with this diagnosis. The patient's sudden-onset severe epigastric pain, radiation to the back, and signs of peritoneal irritation are highly suggestive of gastric perforation, which is a surgical emergency requiring prompt intervention. Perforated appendicitis typically presents with localized right lower quadrant pain that may later become diffuse. While it can cause peritoneal irritation, it is less likely to present with severe epigastric pain and is more commonly associated with focal tenderness in the right lower quadrant. Biliary colic, caused by the passage of a gallstone through the bile ducts, typically presents with episodic, colicky right upper quadrant or epigastric pain. While it can be severe, it does not usually lead to rebound tenderness and guarding unless complicated by gallbladder perforation or cholecystitis.

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What is the initial management for the patient above?

- ☒ Administer intravenous fluids and analgesia, then obtain urgent surgical consultation
- ☐ Initiate PPI therapy and perform endoscopy
- ☐ Obtain laboratory tests including CBC, comprehensive metabolic panel and lipase level
- ☐ Perform urgent abdominal CT scan

**Explanation :**

Given the patient's presentation of sudden-onset severe epigastric pain, which awakened him from sleep, along with signs of peritoneal irritation such as guarding and rebound tenderness, the initial management should focus on stabilizing the patient and obtaining urgent surgical consultation. Intravenous fluids help correct any dehydration or hypovolemia due to fluid loss from potential gastric leakage, and analgesia helps alleviate the patient's pain. However, definitive management of suspected perforated peptic ulcer disease (PUD) requires surgical intervention, hence the need for urgent surgical consultation.

14

A 26-year-old pregnant woman presents to the obstetrics clinic with complaints of lower abdominal pain, vaginal discharge, and dyspareunia. On examination, she has cervical motion tenderness, bilateral adnexal tenderness, and purulent cervical discharge. Ultrasound confirms an intrauterine pregnancy with no signs of complications. What is the management principle for this patient?

- ☐ Outpatient management
- ☒ Admission for inpatient treatment
- ☐ Fluid analysis from cul-de-sac
- ☐ Immediate surgical intervention

**Explanation :**

A lower abdominal pain, vaginal discharge, cervical motion tenderness, bilateral adnexal tenderness, and purulent cervical discharge, consistent with pelvic inflammatory disease (PID), and her pregnancy status, the management principle is admission for inpatient treatment. Outpatient management is not appropriate for pregnant patients with PID due to the risk of complications such as tubo-ovarian abscess formation and adverse pregnancy outcomes. Inpatient treatment allows for close monitoring and administration of parenteral antibiotics. Inpatient treatment is recommended to ensure close monitoring, administration of appropriate antibiotics, and evaluation





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14

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15

A 30-year-old patient presents to the emergency department with sudden-onset intermittent colicky abdominal pain. He describes the pain as severe and crampy, with episodes lasting several minutes and occurring intermittently over the past few hours. On examination, there is no abdominal tenderness, guarding, or rebound tenderness. The patient denies any fever or other associated symptoms. What is the most likely diagnosis?

- ☐ Gangrenous small bowel obstruction
- ☐ Gangrenous large bowel obstruction
- ☒ Simple small bowel obstruction
- ☐ Acute appendicitis

**Explanation :**

Given the patient's presentation of sudden-onset intermittent colicky abdominal pain without tenderness or fever, the most likely diagnosis is simple small bowel obstruction (SBO). SBO presents with sudden-onset colicky abdominal pain, typically due to intermittent bowel peristalsis against an obstructed segment of the small intestine. In the absence of signs of strangulation or ischemia, such as tenderness and fever, the obstruction is likely simple and non-gangrenous. Gangrenous SBO typically presents with severe, constant abdominal pain, associated with tenderness, guarding, and possibly fever. The absence of these findings makes this diagnosis less likely.

Acute appendicitis typically presents with localized right lower quadrant pain, associated with tenderness, guarding, and often fever. The absence of these findings makes acute appendicitis less likely as the cause of the patient's symptoms.

16

The "discriminatory zone" is the quantitative serum  $\beta$ -hCG level at which a normal intrauterine gestation should be seen on ultrasonography. The discriminatory zone for transvaginal





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The "discriminatory zone" is the quantitative serum  $\beta$ -hCG level at which a normal intrauterine gestation should be seen on ultrasonography. The discriminatory zone for transvaginal ultrasonography is:

- ☐ 500 to 1,000 mIU/mL  $\beta$ -hCG
- ☒ 1,000 to 2,000 mIU/mL  $\beta$ -hCG
- ☐ 2,000 to 3,000 mIU/mL  $\beta$ -hCG
- ☐ > 3,000 mIU/mL  $\beta$ -hCG

**Explanation :**

The discriminatory zone is the quantitative serum  $\beta$ -hCG level at which a normal pregnancy can be detected by either transvaginal or transabdominal ultrasonography. As ultrasonographic technology improves, the discriminatory zone continues to drop. Furthermore, these levels may vary somewhat between hospitals due to technologic differences. However, the accepted range is 1,000 to 2,000 mIU/mL for transvaginal ultrasonography, and 2,400 to 3,600 mIU/mL for transabdominal ultrasonography. Other sources cite a range of 1,500 to 2,500 mIU/mL as the accepted range in transabdominal ultrasonography. Of course, the higher the discriminatory threshold, the higher the specificity for an abnormal pregnancy, including ectopic gestation.

17

A 28-year-old patient presents to the emergency department with complaints of passage of liquor per vagina for the past 20 hours. She reports associated abdominal discomfort, foul smelling vaginal discharge and fever. On examination, tachycardia and the patient appear uncomfortable, and there is evidence of fetal distress on fetal monitoring. Abdominal examination reveals tenderness and guarding. What is the most likely diagnosis?

- ☐ Preterm PROM
- ☐ Placental abruption
- ☒ Chorioamnionitis
- ☐ Ectopic pregnancy

**Explanation :**

Chorioamnionitis is an infection of the chorion and amnion, typically occurring following prolonged rupture of membranes. It presents with fever, leukocytosis, abdominal tenderness, maternal discomfort, and evidence of fetal distress. Prompt recognition and treatment are crucial to prevent maternal and fetal complications. Given the patient's presentation of passage of liquor per vagina for 20 hours, fever, leukocytosis, abdominal tenderness, and evidence of fetal distress on fetal monitoring, the most likely diagnosis is chorioamnionitis.

PPROM can lead to the passage of liquor per vagina, it typically presents with a history of rupture of membranes before term gestation without signs of infection. The presence of fever, leukocytosis, and abdominal tenderness suggests an infectious process, making chorioamnionitis a more likely diagnosis.

Placental abruption presents with sudden-onset abdominal pain, vaginal bleeding, and signs of fetal distress. While abdominal tenderness may be present, the absence of significant vaginal bleeding makes placental abruption less likely in this case.

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A 3-day-old newborn presents to the pediatric clinic with jaundice and abnormal body movements. The mother reports that the baby has been feeding poorly and appears more lethargic than usual. On examination, the baby is found to have yellow discoloration of the skin and sclerae. Neurological examination reveals hypertonia and intermittent abnormal movements consistent with seizure-like activity. Laboratory investigations reveal a total bilirubin (BILT) level of 30 mg/dL and a direct bilirubin level of 22 mg/dL. What is the next management step for this patient?

- ☐ Initiate phototherapy
- ☒ Perform exchange transfusion
- ☐ Administer IVIG therapy
- ☐ Obtain urgent brain imaging

#### Explanation :

Given the patient's presentation of jaundice within the first few days of life, with high total and direct bilirubin levels, along with abnormal body movements suggestive of neurological dysfunction, the next management step should be exchange transfusion. Exchange transfusion is indicated in cases of severe hyperbilirubinemia, especially when associated with signs of acute bilirubin encephalopathy or kernicterus. It rapidly reduces serum bilirubin levels and helps prevent neurological sequelae.

Phototherapy is the initial treatment for neonatal jaundice, but in severe cases with neurological complications, such as kernicterus, exchange transfusion is indicated as it rapidly reduces serum bilirubin levels.

IVIG therapy is not indicated for the management of severe hyperbilirubinemia in newborns. It is used in specific cases of hemolytic disease of the newborn due to isoimmune hemolysis.

While brain imaging may be considered in cases of suspected kernicterus to evaluate for neurological damage, exchange transfusion should be initiated urgently to reduce serum bilirubin levels and prevent further neurological injury.

19

A 3-hour-old newborn, born at 33 weeks of gestation, is brought to the neonatal intensive care unit due to fast breathing and failure to effectively suck since birth. The mother reports a history of preterm labor and rupture of membranes for 48 hours before delivery. On examination, the newborn appears tachypneic with nasal flaring and intercostal retractions. Auscultation reveals coarse crackles bilaterally. The baby's oxygen saturation is 90% on room air. What is the most likely diagnosis for this newborn?

- ☐ Respiratory distress syndrome (RDS)
- ☒ Transient tachypnea of the newborn (TTN)
- ☐ Meconium aspiration syndrome (MAS)
- ☐ Pneumonia

#### Explanation :

TTN is caused by delayed clearance of fetal lung fluid, leading to transient respiratory distress shortly after birth. It commonly presents with tachypnea, nasal flaring, intercostal retractions, and mild cyanosis. The history of preterm birth and rupture of membranes for 48 hours is suggestive of TTN. Given the newborn's presentation of fast breathing, failure to suck, tachypnea, and signs of respiratory distress shortly after birth, the most likely diagnosis is transient tachypnea of the newborn (TTN).

RDS is caused by surfactant deficiency in premature infants, typically those born before 37 weeks of gestation. It presents with respiratory distress shortly after birth but is more common in





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**Explanation :**

TTN is caused by delayed clearance of fetal lung fluid, leading to transient respiratory distress shortly after birth. It commonly presents with tachypnea, nasal flaring, intercostal retractions, and mild cyanosis. The history of preterm birth and rupture of membranes for 48 hours is suggestive of TTN. Given the newborn's presentation of fast breathing, failure to suck, tachypnea, and signs of respiratory distress shortly after birth, the most likely diagnosis is transient tachypnea of the newborn (TTN).

RDS is caused by surfactant deficiency in premature infants, typically those born before 37 weeks of gestation. It presents with respiratory distress shortly after birth but is more commonly seen in infants born at earlier gestational ages, such as less than 32 weeks.

MAS occurs when a newborn inhales meconium-stained amniotic fluid, leading to airway obstruction, chemical pneumonitis, and respiratory distress. While MAS can present with similar symptoms, the absence of meconium staining on the skin or umbilical cord and the history of preterm birth make TTN a more likely diagnosis.

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A 6-month-old infant presents to the pediatric emergency department with symptoms suggestive of meningitis. The infant has been irritable, lethargic, and febrile for the past 24 hours. On examination, the fontanelle is tense, and there is neck stiffness. What is the first-line treatment for this infant's suspected meningitis?

- ☒ Ceftriaxone + Vancomycin
- ☐ Ampicillin + Gentamicin
- ☐ Cefotaxime + Ampicillin
- ☐ Vancomycin + Gentamicin

**Explanation :**

Given the presentation of a 6-month-old infant with symptoms suggestive of meningitis, the first-line treatment is empirical antibiotics that cover both Gram-positive and Gram-negative bacteria, including common pathogens associated with meningitis in this age group. Ceftriaxone provides broad coverage against Gram-negative bacteria, including *Neisseria meningitidis* and *Haemophilus influenzae*, while vancomycin covers Gram-positive bacteria, particularly *Streptococcus pneumoniae* and group B *Streptococcus*, which are common causes of meningitis in infants.

While cefotaxime provides broad coverage against Gram-negative bacteria similar to ceftriaxone, ampicillin alone does not provide adequate coverage for common meningitis pathogens in infants. Therefore, this combination is not the preferred first-line treatment for suspected meningitis in this age group.

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A neonate born at 42 weeks of gestation presents with signs and symptoms suggestive of meconium aspiration syndrome (MAS). The infant was born via vaginal delivery with meconium-stained amniotic fluid. On examination, the baby appears cyanotic, with poor respiratory effort, coarse crackles on auscultation, and meconium staining on the skin. What is the appropriate treatment for this neonate with suspected meconium aspiration syndrome?

- ☐ Oxygen therapy and CPAP
- ☒ Endotracheal intubation and mechanical ventilation
- ☐ Intravenous fluids and broad-spectrum antibiotics

☐ Intratracheal instillation of surfactant





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- ☐ Oxygen therapy and CPAP
- ☒ Endotracheal intubation and mechanical ventilation
- ☐ Intravenous fluids and broad-spectrum antibiotics
- ☐ Intratracheal instillation of surfactant

**Explanation :**

Meconium aspiration syndrome (MAS) occurs when a newborn inhales meconium-stained amniotic fluid, leading to airway obstruction, chemical pneumonitis, and respiratory distress. Endotracheal intubation and mechanical ventilation are indicated for infants with MAS who have significant respiratory distress, poor respiratory effort, or persistent hypoxemia despite oxygen therapy. Mechanical ventilation helps to maintain adequate oxygenation and ventilation and may facilitate the clearance of meconium from the airways.

Oxygen therapy and CPAP may be initiated initially to support oxygenation, they may not be sufficient for severe cases of MAS such that the patient has poor respiratory effort. Broad-spectrum antibiotics are not routinely indicated unless there is evidence of concurrent bacterial infection, such as pneumonia. Antibiotics should be selected based on culture results if infection is suspected. Surfactant replacement therapy is effective in treating respiratory distress syndrome (RDS) caused by surfactant deficiency, it is not routinely used in the treatment of MAS.

22

A 10-year-old child presents to the pediatric clinic with intense itching and a rash that has been worsening over the past few weeks. On examination, there are multiple excoriated papules and burrows on the wrists, interdigital spaces, and flexural areas of the body. What is the appropriate treatment for scabies in this patient?

- ☒ Topical permethrin
- ☐ Oral ivermectin
- ☐ Topical corticosteroids
- ☐ Oral antibiotics

**Explanation :**

Scabies is a parasitic infestation caused by the mite *Sarcoptes scabiei*. Treatment aims to eradicate the mites and relieve symptoms. This is the preferred first-line treatment for scabies. Permethrin is a topical insecticide that effectively kills scabies mites and their eggs. It is applied to the entire body from the neck down and left on for 8-14 hours before washing off. Repeat treatment may be necessary after 1-2 weeks.

Oral ivermectin is an alternative treatment for scabies, especially in cases where topical therapy is impractical or ineffective. It is generally reserved for cases that are refractory to topical treatments or for patients who cannot tolerate topical medications. It may also be used in outbreaks or in individuals with crusted scabies. Topical corticosteroids may provide symptomatic relief for the itching associated with scabies, but they do not treat the underlying infestation.

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A 45-year-old man, living on the streets, is brought to the emergency department with complaints of severe headache, muscle aches, and high-grade fever. He reports that these symptoms have been occurring intermittently for the past two weeks. On examination, he is febrile with a temperature of  $39.5^{\circ}\text{C}$  and has splenomegaly, with the spleen palpable 3 cm below the costal margin. He denies any recent travel or known exposures. What is the most likely diagnosis for this patient?

☒ Relapsing fever☐ Typhoid fever☐ Typhus☐ Malaria**Explanation :**

Base on the patient's presentation of recurrent high-grade fever, severe headache, myalgia, and splenomegaly, the most likely diagnosis is relapsing fever. Relapsing fever, caused by *Borrelia* species transmitted by lice or ticks, is characterized by recurrent episodes of fever with asymptomatic intervals. It commonly occurs in individuals living in overcrowded or unsanitary conditions, such as homeless individuals. The presence of splenomegaly is a typical finding in relapsing fever.

Typhoid fever, caused by *Salmonella typhi*, can present with symptoms similar to relapsing fever, including high fever, headache, myalgia, and splenomegaly. However, typhoid fever is more commonly associated with gastrointestinal symptoms such as abdominal pain, diarrhea, and constipation.

Typhus, caused by *Rickettsia* species transmitted by fleas or lice, can present with fever, headache, myalgia, and occasionally splenomegaly. However, the patient's history of recurrent fever episodes over the past two weeks is more suggestive of relapsing fever.

Malaria can cause recurrent episodes of fever, but the absence of travel history to endemic areas and exposure to malaria-carrying mosquitoes makes this diagnosis less likely. Malaria may also present with other symptoms such as chills, rigors, and anemia, which were not described in the patient's presentation.

24

A 21-year-old male with a known history of type 1 diabetes presents to the emergency department with a loss of consciousness for the past 2 hours. En route to the hospital, he experienced two episodes of vomiting. On arrival, he is found to be tachycardic and tachypneic, with deep and rapid breathing. His blood glucose level is 450 mg/dL (25 mmol/L). What is the most likely diagnosis for this patient?

☒ Diabetic ketoacidosis (DKA)☐ Hyperosmolar hyperglycemic state (HHS)☐ Hypoglycemia☐ Hyperglycemic hyperosmolar nonketotic syndrome (HHNS)**Explanation :**

Given the patient's history of type 1 diabetes, along with the presentation of loss of consciousness, vomiting, tachycardia, and deep and rapid breathing, the most likely diagnosis is diabetic ketoacidosis (DKA). DKA is a severe complication of uncontrolled diabetes characterized by hyperglycemia, ketosis, and metabolic acidosis. The patient's elevated blood glucose level, along with the presence of ketones in the urine and metabolic acidosis, support the diagnosis of DKA.

While HHS presents with severe hyperglycemia and dehydration similar to DKA, it typically occurs in





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- ☐ Hyperosmolar hyperglycemic state (HHS)
- ☐ Hypoglycemia
- ☐ Hyperglycemic hyperosmolar nonketotic syndrome (HHNS)

**Explanation :**

Given the patient's history of type 1 diabetes, along with the presentation of loss of consciousness, vomiting, tachycardia, and deep and rapid breathing, the most likely diagnosis is diabetic ketoacidosis (DKA). DKA is a severe complication of uncontrolled diabetes characterized by hyperglycemia, ketosis, and metabolic acidosis. The patient's elevated blood glucose level, along with the presence of ketones in the urine and metabolic acidosis, support the diagnosis of DKA. While HHS presents with severe hyperglycemia and dehydration similar to DKA, it typically occurs in type 2 diabetes and is less common in younger individuals like the patient described. Additionally, HHS is characterized by extreme hyperosmolality without significant ketosis or acidosis, which contrasts with the metabolic acidosis seen in DKA.

Hypoglycemia is unlikely given the patient's markedly elevated blood glucose level. Hypoglycemia typically presents with symptoms such as sweating, tremors, and confusion, which are not described in this case.

HHNS shares similarities with HHS but is characterized by extreme hyperglycemia without significant ketosis or acidosis. However, HHNS is less common in type 1 diabetes and is more commonly seen in older individuals with type 2 diabetes.

25

A 2-year-old male child is brought to the pediatric clinic by his parents due to diarrhea and vomiting persisting for the past 2 days. On examination, he appears lethargic with sunken eyes and decreased breast feeding. His mucous membranes are dry, and he has reduced urine output. What is the most appropriate initial treatment for this child?

☒ Oral rehydration therapy (ORT)

- ☐ Intravenous fluids
- ☐ Antibiotics
- ☐ Antiemetics

**Explanation :**

Given the child's presentation of diarrhea, vomiting, decreased oral intake, lethargy, sunken eyes, and reduced urine output, the most appropriate initial treatment is oral rehydration therapy (ORT). ORT is the cornerstone of treatment for dehydration due to diarrhea and vomiting in children. It involves administering oral rehydration solution (ORS) containing electrolytes and glucose to replace lost fluids and electrolytes. ORT is effective in the majority of cases of mild to moderate dehydration and can prevent the need for hospitalization and intravenous fluids.

Intravenous fluids may be necessary for children with severe dehydration or those who cannot tolerate oral fluids, they are not typically indicated as the initial treatment for dehydration. ORT is preferred whenever possible due to its effectiveness, ease of administration, and lower risk of complications.

Antiemetics may be considered to control vomiting in children with gastroenteritis, but they are not the initial treatment for dehydration. ORT to replace lost fluids and electrolytes is the priority in managing dehydration due to diarrhea and vomiting.





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An 8-month-old infant is brought to the pediatric emergency department with a history of diarrhea and vomiting for the past 4 days. On examination, the infant appears lethargic, with sunken fontanelles and a failure to suck. His extremities are cold, and his pulse is feeble. What is the most likely diagnosis for this infant?

- ☐ Severe dehydration
- ☐ Moderate dehydration
- ☒ Hypovolemic shock
- ☐ Gastroenteritis

**Explanation :**

Hypovolemic shock occurs when there is significant fluid loss leading to inadequate circulating blood volume and compromised tissue perfusion. The infant's clinical presentation of lethargy, failure to suck, cold extremities, and feeble pulse is consistent with hypovolemic shock due to severe dehydration. Given the infant's presentation of diarrhea, vomiting, lethargy, failure to suck, cold extremities, and feeble pulse, the most likely diagnosis is hypovolemic shock. While gastroenteritis may be the underlying cause of the diarrhea and vomiting, the infant's clinical presentation goes beyond the typical symptoms of gastroenteritis and is indicative of severe dehydration progressing to hypovolemic shock.

27

During the waste disposal process at a hospital, the management needs to decide which type of waste can be disposed of in a biogas system. Biogas systems are used to treat organic waste, converting it into renewable energy and bio-fertilizer. Which of the following types of waste can be disposed of in a biogas system?

- ☐ Plastic waste
- ☐ Glass waste
- ☒ Organic food waste
- ☐ Metal waste

**Explanation :**

Proper segregation of waste is crucial in hospitals to ensure that different types of waste are disposed of in the most environmentally friendly and efficient manner. Organic waste, such as food scraps and biodegradable materials, should be separated from non-biodegradable waste like plastics, metals, and glass to optimize waste management processes. Biogas systems utilize anaerobic digestion to break down organic matter, such as food waste, agricultural waste, and manure, to produce biogas (a mixture of methane and carbon dioxide) and digestate (a nutrient-rich fertilizer). Biogas can be used as a renewable energy source for heating, electricity, or as a fuel for vehicles, while the digestate can be used as a bio-fertilizer in agriculture. Organic food waste is biodegradable and can be effectively broken down in a biogas system. The decomposition of organic matter by microorganisms in a biogas digester produces methane and carbon dioxide, which can be used as a source of renewable energy. Plastic, glass and metal waste are not biodegradable and cannot be broken down in a biogas system. Both require specific recycling processes or disposal in landfills, as they do not decompose into organic matter suitable for biogas production.

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A 28-year-old female presents to the emergency department with profuse vaginal bleeding 30 minutes after delivering a healthy baby. The delivery was uncomplicated, but the placenta was delivered using traction without controlled cord traction. The patient appears pale and tachycardic, and her blood pressure is 90/60 mmHg. On physical examination, there is significant uterine bleeding, and the uterus feels firm. What is the most likely cause of the postpartum hemorrhage (PPH) in this patient?

- ☐ Uterine atony
- ☐ Retained placental fragments
- ☒ Uterine inversion
- ☐ Genital tract laceration

**Explanation :**

Uterine Inversion is a rare obstetric emergency where the uterus turns inside out, typically following childbirth. It can result from excessive traction on the umbilical cord, particularly when the uterus is not well contracted. Uterine inversion can occur if excessive traction is applied to the umbilical cord without adequate uterine support, especially if controlled cord traction is not used. This results in the uterus turning inside out, leading to severe hemorrhage. The firm uterus felt on examination could be the inverted uterine fundus palpable through the cervix or vaginal canal. Management includes immediate manual or surgical repositioning of the uterus, stabilization of the patient's hemodynamic status, and administration of uterotonic agents after repositioning to ensure proper uterine contraction.

Controlled Cord Traction (CCT) is a technique used during the third stage of labor to assist with the delivery of the placenta. It involves gently pulling on the umbilical cord while applying counter-pressure to the uterus, reducing the risk of uterine inversion and other complications associated with manual extraction of the placenta. Proper training and technique are crucial in preventing adverse outcomes.

Uterine atony is the most common cause of PPH, characterized by a soft and poorly contracted uterus. However, in this case, the uterus feels firm on examination, making uterine atony less likely. Lacerations of the genital tract can also cause significant bleeding, but they are less likely to present with a firm uterus and would typically be associated with a traumatic delivery or instrumental delivery.

29

A 24-year-old primigravida at 28 weeks of gestation presents with complaints of painless vaginal bleeding that started two hours ago. She has no abdominal pain or contractions. Her prenatal course has been uncomplicated. On examination, her vital signs are stable, and the fetal heart rate is 140 beats per minute. Speculum examination reveals moderate bleeding from the cervical os. An ultrasound performed in the emergency department shows a placenta completely covering the internal cervical os. What is the most appropriate management at this time?

- ☐ Immediate cesarean section
- ☒ Administration of corticosteroids and observation
- ☐ Digital cervical examination
- ☐ Discharge home with follow-up in one week

**Explanation :**

Administering corticosteroids helps in fetal lung maturity, and careful observation is necessary given the risk of recurrent bleeding in placenta previa.





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- ☐ Digital cervical examination
- ☐ Discharge home with follow-up in one week

**Explanation :**

Administering corticosteroids helps in fetal lung maturity, and careful observation is necessary given the risk of recurrent bleeding in placenta previa. Immediate cesarean section might be necessary if there were signs of maternal or fetal distress or massive hemorrhage, the patient is currently stable. Digital cervical examination is contraindicated in cases of placenta previa due to the risk of exacerbating the bleeding.

30

A 28-year-old woman, gravida 1 para 1, presents to the emergency department 12 hours after delivering a healthy baby. She reports experiencing a sudden seizure that lasted approximately 2 minutes. Her husband confirms that she has no prior history of epilepsy or seizure disorders. On examination, her blood pressure is 160/100 mmHg, and she has generalized edema. There are no signs of infection or trauma. What is the most likely diagnosis?

- ☒ Eclampsia
- ☐ Epilepsy
- ☐ Postpartum psychosis
- ☐ Hypocalcemia

**Explanation :**

Eclampsia is a severe complication of preeclampsia characterized by the onset of seizures in a woman with preeclampsia. The presence of hypertension (BP 160/100 mmHg), generalized edema, and the occurrence of a seizure 12 hours postpartum strongly suggest eclampsia. Eclampsia can occur during pregnancy, labor, or postpartum. The management of eclampsia includes:

- Stabilization (ABCD)
- Administering magnesium sulfate to prevent further seizures.
- Antihypertensive Therapy (labetalol or hydralazine)
- Monitoring

31

A 43-year-old woman presents to the ED with a 3-week history of intermittent headache, nausea, and fatigue. She was seen at her private doctor's office 1 week ago along with her husband and children, who also have similar symptoms. They were diagnosed with a viral syndrome and told to increase their fluid intake. She states that the symptoms began approximately when it started to get cold outside. The symptoms are worse in the morning and improve while she is at work. Her





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- ☐ Order a Mono spot test
- ☐ Perform a nasal pharyngeal swab to test for influenza
- ☐ Consult psychiatry to evaluate for malingering

☒ Order a carboxyhemoglobin (COHb) level

**Explanation :**

The most useful diagnostic test in suspected CO poisoning is a COHb level. Normal levels are less than 5% in nonsmokers as CO is a natural byproduct of the metabolism of porphyrins. COHb levels may be slightly more elevated in smokers and those who live in large cities. CO poisoning should be suspected when multiple patients (usually in the same family) present with flu-like symptoms, and were exposed to products of combustion (ie, home heaters/generators). The level may not be elevated if they have not been exposed recently (<12 hours). This most commonly occurs in colder, winter months. The mainstay of treatment is the delivery of oxygen. Hyperbaric oxygen is usually used for patients with COHb levels greater than 25%. The patient's house should be evaluated by the fire department or a heating and cooling specialist.

A Mono spot test can be helpful in detecting acute mononucleosis. Without pharyngitis, lymphadenopathy and given the duration of symptoms, this choice is not likely. CO poisoning is often confused for a viral syndrome. Patients with influenza usually present to the ED with high fever. The duration of symptoms also makes this less likely. Malingering is the intentional production of false or exaggerated symptoms motivated by external incentives. It is unlikely to happen in entire families. Lead toxicity is mainly a disease of children resulting from ingestion of lead-based paints. Adults can be exposed to lead in a variety of occupational circumstances, such as welders, glassmakers, and scrap metal workers. There is no classic presentation of lead toxicity. Therefore, high suspicion and a thorough history are critical. The diagnosis is made by an elevated whole-blood lead level.

32

A 47-year-old man is brought to the ED by EMS after being found wandering in the street mumbling. His BP is 150/75 mm Hg, HR is 110 beats/minute, RR is 16 breaths/minute, temperature is 100.5°F, oxygen saturation is 99% on room air, and point-of-care glucose is 98 mg/dL. On examination, the patient is confused with mumbling speech. His pupils are dilated and face is flushed. His mucous membranes and skin are dry. Which of the following toxic syndromes is this patient exhibiting?

- ☐ Sympathomimetic syndrome

☒ Anticholinergic syndrome

- ☐ Cholinergic syndrome
- ☐ Opioid intoxication

**Explanation :**

The term toxidrome refers to a constellation of physical findings that can provide important clues





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- ☐ Sympathomimetic syndrome
- ☒ Anticholinergic syndrome
- ☐ Cholinergic syndrome
- ☐ Opioid intoxication

**Explanation :**

The term toxidrome refers to a constellation of physical findings that can provide important clues in a toxic ingestion. This is particularly useful in patients who cannot provide an adequate history. The anticholinergic syndrome typically presents with delirium, mumbling speech, tachycardia, elevated temperature, flushed face, dry mucous membranes and skin, dilated pupils, and hypoactive bowel sounds. The anticholinergic syndrome can be remembered by the phrase "blind as a bat (mydriasis), red as a beet (flushed skin), hot as a hare (hyperthermia secondary to lack of sweating), dry as a bone (dry mucous membranes), and mad as a hatter (mental status changes)." The sympathomimetic syndrome is usually seen after ingestion of cocaine, amphetamines, or decongestants. It typically presents with delirium, paranoia, tachycardia, hypertension, hyperpyrexia, diaphoresis, mydriasis, seizures, and hyperactive bowel sounds. Sympathomimetic and anticholinergic syndromes are frequently difficult to distinguish. The main difference is that sympathomimetics usually cause diaphoresis whereas anticholinergics cause dry skin. The cholinergic syndrome is commonly remembered by the mnemonics SLUDGE or DUMBBELS. Opioids and ethanol are part of the sedative-hypnotic syndrome. It typically presents with sedation, miosis, respiratory depression, hypotension, bradycardia, hypothermia, and decreased bowel sounds.

33

A 55-year-old male patient presents to the emergency department with a history of vomiting bright red blood for the past two hours. On arrival, his blood pressure is 90/60 mm Hg, and his heart rate is 110 beats per minute. The patient appears pale and diaphoretic. He was resuscitated by the nurse with 2 liters of normal saline. Despite fluid resuscitation, his blood pressure remains low and he continues to show signs of hypovolemic shock. Laboratory results are pending. What is the next best step in the management of this patient?

- ☒ Blood transfusion
- ☐ Intravenous omeprazole
- ☐ Endoscopic evaluation
- ☐ Administration of vasopressors

**Explanation :**

This patient is presenting with signs of hypovolemic shock due to significant upper gastrointestinal (GI) bleeding, as evidenced by his low blood pressure and tachycardia despite initial fluid resuscitation. Blood transfusion is necessary to stabilize the patient hemodynamically and to replace the lost blood volume. Immediate transfusion is crucial in patients with significant blood loss to prevent further deterioration and to ensure adequate oxygen delivery to tissues. Endoscopic evaluation is essential for diagnosing the source of bleeding and potentially treating it (e.g., through banding or cauterization) but it should be performed after initial stabilization. In this





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Endoscopic evaluation is essential for diagnosing the source of bleeding and potentially treating it (e.g., through banding or cauterization), but it should be performed after initial stabilization. In this case, blood transfusion to stabilize the patient's hemodynamics takes precedence over endoscopy. Vasopressors can be considered if the patient remains hypotensive despite adequate fluid and blood resuscitation. However, the initial step should be volume replacement with blood transfusion, as the primary issue is hypovolemia due to blood loss.

34

A 28-year-old G4P3 woman at 36 weeks gestational age presents with sudden onset, severe abdominal pain associated with vaginal bleeding after she was involved in a motor vehicle collision (MVC). She denies associated uterine contractions. She reports loss of fetal movements shortly after the onset of pain. All three of her previous children were born via cesarean section. Her vitals are significant for a BP of 70/40 mm Hg and an HR of 140 beats/minute. Which of the following is the most likely diagnosis?

- ☒ Uterine rupture
- ☐ Premature labor
- ☐ Placenta accreta
- ☐ Placenta previa

**Explanation :**

Uterine rupture is a rare and often catastrophic complication of pregnancy that manifests variably with abdominal pain, loss of uterine contractions (when rupture occurs during labor), vaginal bleeding, and maternal hemorrhagic shock, but may only be initially evident by signs of fetal distress. The most important risk factor is previous cesarean section. Other risk factors include congenital uterine anomalies, multiparity, previous uterine instrumentation, and abdominal trauma (such as the case in this scenario). This is an obstetric emergency that necessitates emergent surgical intervention (cesarean section) for stabilization of both mother and child. Premature labor is not likely to present with hemodynamic instability and loss of uterine contractions. Placenta accreta occurs when the placenta abnormally implants into the uterus and invades deep into the uterine wall. This is a potentially life-threatening cause of maternal hemorrhage that usually manifests as painless vaginal bleeding following delivery. Placenta previa occurs when the placenta partially or completely overlies the cervical os and usually presents with painless vaginal bleeding in the second trimester of pregnancy.

35

What is the most appropriate next step in her management?

- ☒ Establish IV access and begin fluid resuscitation
- ☐ Check fetal heart tones with bedside ultrasound
- ☐ Perform a speculum examination
- ☐ Administer intramuscular morphine

**Explanation :**

The patient is hemodynamically unstable and likely developing hemorrhagic shock, therefore large-bore IV access should be established and she requires immediate fluid resuscitation with crystalloid and likely will ultimately require transfusion of blood products. Consideration of patient positioning (elevating the patient with pillows 15 degrees onto the left lateral decubitus position) may also improve BP as it will prevent the uterus from compressing the inferior vena cava. In this case, strict spinal precautions should be considered when manipulating the patient.





35

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36

A 50-year-old known patient with rheumatic valvular heart disease presents to the emergency department with signs of shock. On examination, he is hypotensive with a blood pressure of 80/50 mmHg, tachycardic with a heart rate of 120 beats per minute, and tachypneic with a respiratory rate of 30 breaths per minute. His skin is cool and clammy, and he appears lethargic. Initial resuscitation measures, including the administration of 3 liters of crystalloid fluid, have been initiated. What is the next best step in management?

- ☒ Initiate noradrenaline infusion
- ☐ Administer colloid fluid
- ☐ Reassess fluid status
- ☐ Administer additional crystalloid fluid bolus

**Explanation :**

In patients with shock who do not respond adequately to initial fluid resuscitation, vasopressor therapy such as noradrenaline is indicated. Noradrenaline is a potent vasopressor that acts on alpha-adrenergic receptors to increase systemic vascular resistance, leading to an increase in blood pressure, thus helping to restore blood pressure in hypotensive patients. Noradrenaline is considered the first-line vasopressor for patients with distributive shock and is recommended in international guidelines for the management of septic shock. Administering additional crystalloid fluid bolus This may exacerbate fluid overload and is not the appropriate next step after the patient has already received 2 liters of crystalloid fluid without improvement in blood pressure. Vasopressor therapy is indicated at this stage.

Colloid fluids may be considered in some cases of shock, such as those with severe hypoalbuminemia or when crystalloid fluid resuscitation has been ineffective, the first-line vasopressor of choice for patients with distributive shock like septic shock is noradrenaline.





37

A 58-year-old man presents to the emergency department (ED) with blister formation on both feet that he first noticed 2 days ago. He denies past medical history, medication use, or drug allergies. His social history is significant for alcohol dependence and he recently became homeless. He denies any sick contacts or recent travel. Upon physical examination, the lesions are fluid-filled. His feet are grossly cyanotic and tender to the touch. Pulses are present by Doppler. His foot is shown below. What is the most likely diagnosis?

- ☐ Chilblains
- ☒ Frostbite
- ☐ Trench foot
- ☐ Thermal burn

**Explanation :**

Frostbite usually occurs when tissue temperature falls below  $0^{\circ}\text{C}$  ( $32^{\circ}\text{F}$ ). There are three phases to the freezing injury cascade. Phase 1 (prefreeze, without actual ice formation) occurs when the tissue reaches a temperature below  $10^{\circ}\text{C}$  and includes initial skin cooling, increased blood viscosity, vasoconstriction, and microvascular leakage that causes localized edema formation. Patients typically lose cutaneous sensation in the tissue during this phase. Phase 2 (freeze-thaw) occurs when extracellular crystal formation begins, thereby causing intracellular shrinkage, protein and lipid derangements, cell dehydration, and collapse of the cellular network. Finally, phase 3 (vascular stasis and progressive ischemia) begins once the tissue is rewarmed and involves further coagulation, interstitial leakage, cytokine release, and cell death. This results in blister formation, cyanosis, and ultimately mummification of the tissue. The bullae formed may also have a hemorrhagic appearance, but not always. It is important to note that wind and moisture may increase the freezing rate. Management includes rapid rewarming in a circulating water bath with water temperatures of  $37^{\circ}\text{C}$  to  $39^{\circ}\text{C}$ . Tissue massage, which furthers tissue loss, should be avoided as should dry heat sources, such as a space heater. Rewarming is a painful procedure that requires parenteral analgesia. Patients may also have a degree of dehydration and benefit from warmed crystalloid administration.

Chilblains, also known as pernio, is an abnormal vascular response to cold resulting in erythema, itching, and inflammation of the skin. Pernio is commonly seen in the homeless population as a result of chronic dry-cold exposure and mostly affects the face, hands, and pretibial areas, particularly in young women or those with autoimmune disorders. Trench foot, also known as immersion injury, is also common in the homeless population. This occurs when skin is exposed chronically to wet and cold environments; however, the ambient temperature is above freezing. It usually presents as a loss of sensation with pallor and mottled skin that is sodden and friable. Thermal burns may present with bullae formation but do not elicit cyanosis and this patient's history is not consistent with thermal injury.

38

The director of a school notices that the noise from an old air conditioner is disturbing students during classes. They decide to take action to address the issue. What is the best option for the director to solve the problem?

- ☒ Replace the old air conditioner
- ☐ Build a noise protective wall around the air conditioner
- ☐ Shift the students to a different classroom
- ☐ Provide earplugs to the students



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**Explanation :**

Replacing the old air conditioner with a new, quieter model addresses the root cause of the noise disturbance. Newer models of air conditioners are designed to operate more quietly, reducing noise levels in the classroom and creating a better learning environment for students. Building a noise protective wall may help reduce the transmission of noise to some extent, it does not address the underlying issue of the noisy air conditioner itself. Additionally, constructing a wall may be a costly and time-consuming solution compared to simply replacing the air conditioner. Importance of Addressing the Root Cause: In situations like this, addressing the root cause of the problem, such as the noisy air conditioner, is essential for creating a conducive learning environment for students.

39

A 60-year-old known diabetic patient presents with a gradual onset of bilateral decreased vision over the past six months. He reports experiencing difficulty reading and driving, with no improvement even with corrective lenses. On physical examination, there is bilateral cloudiness observed in the pupils, along with a white reflex upon direct ophthalmoscopic examination. What is the most likely diagnosis in this patient?

- ☒ Diabetic cataracts
- ☐ Diabetic retinopathy
- ☐ Macular degeneration
- ☐ Glaucoma

**Explanation :**

Based on the patient's history of diabetes and the presence of bilateral cloudiness and a white reflex in the pupils, the most likely diagnosis is diabetic cataracts. Diabetic cataracts can develop earlier and progress more rapidly in diabetic patients compared to the general population. Treatment for cataracts may involve surgical removal of the cloudy lens and replacement with an artificial intraocular lens. Regular eye examinations are essential for diabetic patients to detect and monitor diabetic eye complications such as cataracts and diabetic retinopathy. Early detection and timely intervention can help prevent vision loss and improve outcomes in diabetic patients. Regular eye examinations, along with optimal management of diabetes and other comorbidities, are crucial for preserving vision and overall eye health. Diabetic retinopathy is a common complication of diabetes characterized by damage to the blood vessels of the retina. While diabetic retinopathy can cause vision loss, it typically presents with findings such as microaneurysms, hemorrhages, and exudates in the retina, rather than the cloudiness seen in cataracts.

Glaucoma is a group of eye diseases characterized by damage to the optic nerve, often associated with increased intraocular pressure. While glaucoma can lead to vision loss, it typically presents





with increased intraocular pressure. While glaucoma can lead to vision loss, it typically presents with optic disc cupping and visual field defects rather than cloudiness in the pupils.

40

A 30-year-old man with type 1 DM presents with BP of 100/70 mm Hg and HR is 140 beats/minute. His blood glucose is 750 mg/dL, potassium level is 2.9 mEq/L, bicarbonate is 5 mEq/L, and arterial pH is 7.1. His urine is positive for ketones. Which of the following is the best initial therapy for this patient?

- ☐ Administer a 2 L normal saline bolus and 20 units of subcutaneous insulin
- ☐ Administer 2 ampules of IV sodium bicarbonate and 10 units of IV insulin
- ☒ Administer a 2 L NS bolus and 40 mEq IV potassium chloride
- ☐ Administer a 2 L NS bolus followed by an insulin drip at 0.1 units/kg/hour

#### Explanation :

While the mainstay of treatment for DKA is aggressive IV fluid resuscitation and IV insulin therapy, this patient's serum potassium level is markedly low. He requires fluid resuscitation and potassium correction. If the initial serum potassium is below 3.3 mEq/L, IV KCl should be given prior to administration of insulin. Failure to correct hypokalemia prior to insulin administration could cause respiratory muscle weakness, cardiac arrhythmias, or even cardiac arrest due to intracellular movement of potassium. Once the serum potassium is between 3.3 and 5.3 mEq/L, insulin infusion at 0.1 units/kg/hour is indicated along with IV KCl (20-30 mEq) is added to each liter of IV replacement fluid and continued until the serum potassium concentration has increased to the normal range of 4 to 5 mEq/L.

In DKA, the average adult has a water deficit of 5 to 10 L. After the first 2 L of fluid, regular insulin is administered at a rate of 0.1 units/kg/hour. Insulin must be administered for ketosis and acidosis to resolve. There is no consensus on need for an insulin bolus prior to starting an infusion. Such a bolus may negatively affect the already total body deficit of potassium and in recent years has fallen out of favor in the initial treatment of DKA. Intramuscular and SQ insulin administration is avoided in DKA as absorption may be erratic secondary to volume depletion and poor perfusion. Currently, no study shows a benefit of using bicarbonate in DKA. Bicarbonate administration can cause worsening hypokalemia, paradoxical central nervous system (CNS) acidosis, impaired oxyhemoglobin dissociation, hypertonicity, and sodium overload.

41

A 40-year-old man presents to the ED with a left foot puncture wound from stepping on a rusty nail. His last tetanus vaccine was during childhood and resulted in a severe allergic reaction and a prolonged hospital stay. Which of the following is true regarding tetanus immunization in this patient?

- ☐ Give tetanus toxoid
- ☒ Give tetanus immunoglobulin
- ☐ Give tetanus immunoglobulin and toxoid
- ☐ Give tetanus toxoid and immunoglobulin with diphenhydramine

#### Explanation :

Tetanus toxoid is contraindicated in patients who have had a severe allergic reaction, including respiratory distress or cardiovascular collapse. Tetanus immunoglobulin can be used in these patients. Local reaction at injection site resulting in redness, pain, and swelling is not considered a contraindication.





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41

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The use of tetanus toxoid is contraindicated even if the patient is given diphenhydramine. The patient should be given immunoglobulin, if the injury is high-risk (eg, puncture wound from a rusty nail).

42

An unconscious 51-year-old woman is brought to the emergency medical services. A friend states that the patient was complaining of feeling weak. She vomited and subsequently "blacked out" in the ambulance. The friend states that the patient has no medical problems and takes no medications. She also states that the patient smokes cigarettes and uses cocaine, and that they were snorting cocaine together prior to her blacking out. The patient's BP is 195/80 mm Hg, HR is 50 beats/minute, RR is 7 breaths/minute, and temperature is 98.6°F. What is the eponym associated with her vital signs?

- ☐ Cushing's syndrome
- ☒ Cushing's reflex
- ☐ Cullen's sign
- ☐ Chvostek's sign

**Explanation :**

The patient has a triad of hypertension, bradycardia, and respiratory depression, which is called Cushing's reflex. This is observed in one-third of patients with a potentially lethal increase in ICP. Increased ICP may result from traumatic brain injury or, as in this patient's case, from hemorrhagic stroke and subsequent brain edema. Tobacco and cocaine use are known risk factors for hemorrhagic stroke. Increasing ICP can result in cerebral herniation, which has a mortality rate close to 100%. For any chance of survival, it must be rapidly controlled by intubation, elevation of the head of the bed, hyperventilation, mannitol, and definitive neurosurgical intervention. Cushing's syndrome describes the hyperadrenal state associated with increased production of cortisol, leading to hypertension, truncal obesity, abdominal striae, and hirsutism. Cullen's sign is purplish discoloration around the umbilicus that results from intraperitoneal hemorrhage. Chvostek's sign, associated with hypocalcemia, is twitching of the nose or lips with tapping of the facial nerve.

43





43

A 32-year-old woman presents with a complaint of poor cesarean section wound healing and failure of lactation. The patient had an unremarkable pregnancy and delivered a full-term neonate 2 weeks ago. Her delivery was complicated by maternal hemorrhage and severe hypotension. Her BP is 120/80 mm Hg, HR is 70 beats/minute, RR is 14 breaths/minute, and temperature is 36.6°C. What is the most likely diagnosis?

- ☐ Endometritis
- ☐ Hypothyroidism
- ☒ Sheehan's syndrome
- ☐ Improper lactation technique

**Explanation :**

Sheehan's syndrome is a rare acute dysfunction of hypothalamic-pituitary axis, which may be seen after obstetric hemorrhage and maternal hypotension during delivery. The anterior pituitary gland is especially susceptible to ischemia secondary to maternal hypotension resulting in most commonly prolactin and growth hormone deficiencies. Clinically, it is manifested by failure of lactation, failure of wound healing, and generalized weakness. Normally, levels of prolactin steadily rise during pregnancy in response to sleep, meals, and increase in thyrotropin-releasing hormone (TRH) produced by hypothalamus. IV TRH may be administered and prolactin can be measured before and 30 minutes after TRH administration in order to diagnose Sheehan's syndrome. Normally, levels of prolactin rise at least threefold following administration of TRH. Failure of prolactin to rise in response to TRH infusion is diagnostic of Sheehan's syndrome.

Endometritis would present with signs of systemic illness. Hypothyroidism complicates 0.5% of all pregnancies and is most commonly seen with a preexisting Hashimoto's disease. However, isolated decreased thyroid hormone production would not explain all of the presenting symptoms in this patient. Improper lactation technique may present with failure of lactation, but would not cause generalized weakness or impaired wound healing.

44

A 3-year-old boy presents to the pediatric urologist for follow-up after undergoing orchiopexy six months ago to correct cryptorchidism. The child was diagnosed with cryptorchidism at birth, and the undescended testis was successfully brought down into the scrotum during the surgery. The parents are concerned about the potential long-term health implications for their child. Which of the following conditions is this child at increased risk for in the future despite successful surgical correction?

- ☒ Testicular malignancy
- ☐ Infertility
- ☐ Testicular torsion
- ☐ Inguinal hernia

**Explanation :**

Cryptorchidism is a condition where one or both of the testes fail to descend into the scrotum. It is the most common congenital abnormality of the genitourinary tract in male infants. Early diagnosis and treatment, typically by surgical correction (orchiopexy), are important to reduce the risks of infertility and malignancy. Orchiopexy is surgical procedure is performed to move an undescended testicle into the scrotum and permanently fix it there. It is typically done between 6 months and 2 years of age to optimize outcomes regarding fertility and malignancy risk. Even after successful surgical correction of cryptorchidism, individuals remain at an increased risk





and decreases if orchiopexy is performed before puberty, but it never completely returns to the baseline risk of the general population. Infertility is also a potential risk associated with cryptorchidism, especially if the condition is bilateral and surgery is delayed, the primary concern remains the increased risk of testicular malignancy. Early surgical correction can improve fertility outcomes, but the risk of cancer persists. Testicular torsion is a condition where the spermatic cord twists, cutting off the blood supply to the testicle. Although individuals with undescended testicles are at a slightly higher risk of torsion before surgery, this risk is significantly reduced after orchiopexy. Therefore, it is not the most significant long-term risk post-surgery. Cryptorchidism is often associated with an increased risk of inguinal hernia, but this risk is generally addressed and mitigated during the orchiopexy procedure. It is not considered the primary long-term risk after successful surgery.

45

A 74-year-old woman is brought to the ED by EMS from a nursing home for altered mental status. She is currently being treated for breast cancer. Her BP is 138/72 mm Hg, HR is 90 beats/minute, RR is 17 breaths/minute, and temperature is 38.7°C. A head CT is normal. LP results revealed the following: WBC count: 1020/mm<sup>3</sup> with 90% PMNs, Glucose: 21 mg/dL Protein: 225 g/L. Which of the following is the most likely diagnosis?

- ☐ TB meningitis
- ☒ Bacterial meningitis
- ☐ Viral meningitis
- ☐ Fungal meningitis

#### Explanation :

The CSF analysis in bacterial meningitis typically shows an elevated WBC count with predominant PMNs. Protein is elevated and glucose is low. A Gram stain may show bacteria. The most specific marker for the diagnosis is a positive culture. Tests that evaluate the presence of antigen in the CSF (ie, latex agglutination studies) are particularly useful in the diagnosis of partially treated bacterial meningitis.

Tuberculosis meningitis typically presents with less than 1000 WBC/ $\mu$ L with monocytic predominance. Symptoms are generally insidious in nature, typically not appearing until after a week has passed. In viral meningitis (c), the CSF WBC count is made up of lymphocytes or monocytes, but early in the disease PMNs may predominate. CSF glucose is normal and protein is elevated in viral meningitis. A Gram stain will be negative and culture will show no growth. Fungal meningitis is rare and generally presents in immunocompromised patients. LP usually reveals less than 500/ $\mu$ L WBCs.

| Test       | Normal   | Bacterial  | Viral    | Fungal   | TB                            |
|------------|----------|------------|----------|----------|-------------------------------|
| Protein    | <50      | >200       | <200     | >200     | >200                          |
| Glucose    | >40      | <40        | >40      | <40      | <40                           |
| WBCs       | <5       | >1000      | <1000    | <500     | <1000                         |
| Cell type  | Monos    | >50% Polys | Monos    | Monos    | Monos                         |
| Gram stain | Negative | Positive   | Negative | Negative | Positive (Acid-fast bacillus) |

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Which antibiotic regimen should be initiated?

- ☐ Vancomycin, metronidazole and ceftriaxone
- ☒ Vancomycin, ceftriaxone, ampicillin
- ☐ Vancomycin
- ☐ Ceftriaxone and valacyclovir

**Explanation :**

Bacterial meningitis is caused by different organisms in different age groups. Therefore, the empiric antibiotic therapy for suspected bacterial meningitis varies based on age. This older (>50-year-old) woman, who is immunocompromised and resides in a nursing home, needs coverage for *Listeria*, *Neisseria meningitidis*, and *Staphylococcus aureus*. Vancomycin covers methicillin-resistant *Staphylococcus aureus* (MRSA), ampicillin covers *Listeria*, and ceftriaxone is the best option to cover *Neisseria*. This regimen is also acceptable for neonates, who need coverage for Group B streptococcus (GBS).

Vancomycin only or vancomycin and ceftriaxone will not cover *Listeria*, which is needed in this elderly, immunocompromised patient. Valacyclovir would be indicated if the patient was suspected of having Herpes simplex, which is not indicated by this clinical presentation and LP results.

47

A 2-year-old child is brought to the pediatric clinic by his mother due to a pruritic rash that has been present for several weeks. The rash involves the extensor surfaces of the arms and legs and is characterized by dry, red, and scaly patches. The mother notes that the rash spares the diaper area and that the child seems particularly itchy at night. There is no significant past medical history, and the child is otherwise healthy. The family has no pets, and there have been no recent changes in soaps, detergents, or foods. What is the most likely diagnosis?

- ☒ Atopic dermatitis
- ☐ Contact dermatitis
- ☐ Psoriasis
- ☐ Seborrheic dermatitis

**Explanation :**

Atopic dermatitis, also known as eczema, commonly presents in young children with pruritic, red, and scaly lesions on the extensor surfaces of the arms and legs, and often spares the diaper area. It is a chronic inflammatory skin condition. It is associated with a personal or family history of atopy, including asthma, allergic rhinitis, and food allergies. The rash is typically pruritic and may involve the face, scalp, and extensor surfaces in infants and young children. The chronic and relapsing nature of the rash, along with a tendency to worsen at night due to itching, fits well with atopic dermatitis. As the child grows older, the rash often shifts to the flexural areas (antecubital and popliteal fossae). Treatment involves moisturizing the skin to maintain the skin barrier, using topical corticosteroids to reduce inflammation during flares, and addressing potential triggers such as allergens and irritants. Antihistamines may be used to help with pruritus. Many children with atopic dermatitis improve as they get older, although some may continue to have symptoms into adulthood. Education on skincare and avoidance of triggers is essential in managing this condition. Contact dermatitis is a type of rash caused by exposure to an irritant or allergen. It typically presents with well-demarcated areas of redness and sometimes blistering, localized to the area of contact. The absence of a clear history of exposure and the distribution of the rash make this





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Psoriasis can present with red, scaly plaques, but it more commonly affects the scalp, elbows, knees, and the sacral area rather than the extensor surfaces in a young child. Psoriasis also tends to present with well-demarcated plaques with silvery scales, which is not described in this case. Seborrheic dermatitis is characterized by greasy, scaly, yellowish patches, often affecting areas rich in sebaceous glands such as the scalp (cradle cap in infants), face, and sometimes the diaper area. The distribution and appearance described in this case do not match seborrheic dermatitis.

48

A doctor who is in the last hour of his duty shift, performs an emergency procedure for a patient presenting with an incomplete abortion. After completing the procedure, Dr. Bicheger leaves the room and instructs the nurse to inform the patient about the procedure and post-procedure care. Dr. Bicheger then leaves the hospital. Shortly afterward, Dr. Lemene, the next doctor on duty, arrives and takes over care for the patient. What was the mistake in handling the situation?

- ☒ Dr. Bicheger should have personally informed the patient about the procedure before leaving
- ☐ Dr. Lemene should have informed the patient about the procedure upon arrival
- ☐ The nurse should have been responsible for informing the patient about the procedure
- ☐ There was no mistake; the procedure was handled appropriately

**Explanation :**

Dr. Bicheger should have personally informed the patient about the procedure before leaving. It is the responsibility of the physician who performs a procedure to inform the patient about what was done, any findings, potential complications, and post-procedure care. This ensures that the patient receives accurate and comprehensive information directly from the healthcare provider responsible for their care at that moment. This also fosters trust and communication between the patient and physician.

**Informed Consent:** Before performing any medical procedure, it is essential to obtain informed consent from the patient. This includes explaining the nature of the procedure, potential risks, benefits, and alternatives.

**Post-Procedure Communication:** After completing a procedure, the physician should provide the patient with details about what was done, any findings, immediate care instructions, potential complications, and follow-up plans. This should be done directly by the physician who performed the procedure to ensure clear and accurate communication.

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A 32-year-old female veteran presents to the mental health clinic with symptoms of post-traumatic stress disorder (PTSD) following a combat deployment. She reports frequent nightmares, flashbacks, and severe anxiety triggered by loud noises. She avoids crowds and has become increasingly isolated, affecting her daily life and relationships. She has no significant medical history and is not currently taking any medications. Which of the following is the treatment of choice for PTSD in this patient?

- ☒ Fluoxetine
- ☐ Haloperidol
- ☐ Risperidone
- ☐ Lorazepam

**Explanation :**

The management of PTSD often includes a combination of pharmacotherapy and psychotherapy. Cognitive-behavioral therapy (CBT), specifically trauma-focused CBT, is highly effective in treating PTSD. It is important to tailor treatment to the individual patient's needs and to monitor for any side effects or changes in symptomatology. Regular follow-up and support are crucial in managing PTSD effectively. Fluoxetine, a selective serotonin reuptake inhibitor (SSRI), is commonly used as a first-line treatment for PTSD. SSRIs like fluoxetine are effective in reducing symptoms of anxiety, depression, and intrusive thoughts associated with PTSD. They have a well-established safety profile and are generally well-tolerated by patients.

Haloperidol is an antipsychotic medication primarily used to treat schizophrenia and acute psychosis. It is not typically used as a first-line treatment for PTSD due to its side effect profile and lack of evidence supporting its efficacy in treating PTSD symptoms.

Risperidone, an atypical antipsychotic, may be used as an adjunctive treatment in certain cases of PTSD, particularly for patients with severe symptoms or those who do not respond to SSRIs alone, it is not considered a first-line treatment. Antipsychotics are generally reserved for more complex or refractory cases due to their potential side effects.

Lorazepam, a benzodiazepine, may be used short-term for acute anxiety relief but is not suitable for long-term management of PTSD due to the risk of dependence and tolerance. Benzodiazepines do not address the core symptoms of PTSD and are not recommended as a primary treatment option.

50

A 19-year-old woman presents to the clinic with a complaint of vaginal discharge. She reports having her first sexual intercourse one year ago. She has had two sexual partners since then and does not consistently use condoms. She has no significant medical history and is otherwise healthy. Which of the following management options is the least important in this patient?

- ☐ HIV testing
- ☒ Cervical cancer screening
- ☐ Testing for *Neisseria gonorrhoeae* from the discharge
- ☐ Testing for *Chlamydia trachomatis* from the discharge

**Explanation :**

Cervical cancer screening is the least important in this patient because current guidelines recommend starting cervical cancer screening at age 21, regardless of sexual activity. Therefore, it is not immediately necessary for a 19-year-old woman.

HIV testing is important because the patient is sexually active with multiple partners and inconsistent condom use, putting her at risk for HIV. Early detection of HIV is crucial for





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HIV testing is important because the patient is sexually active with multiple partners and inconsistent condom use, putting her at risk for HIV. Early detection of HIV is crucial for management and preventing transmission.

Testing for *Neisseria gonorrhoeae* and *Chlamydia trachomatis* from the discharge is important because the patient's symptoms and sexual history put her at risk for sexually transmitted infections (STIs), including gonorrhea, which can cause vaginal discharge.